

Full Stack Documentation (FSD)

Project Documentation

Project: ShopEEZ | Date: June 2026 | Stack: MongoDB · Express · React · Node.js

1. Project Overview

Field	Details
Project Name	ShopEEZ
Tech Stack	MongoDB, Express.js, React 19, Node.js (MERN)
Authentication	JWT Bearer Token + Bcrypt password hashing
State Management	Redux Toolkit (cartSlice, authSlice, productSlice, wishlistSlice)
Database	MongoDB Atlas (Cloud)
Frontend Port	5173 (Vite Dev Server)
Backend Port	5000 (Express.js)
Currency	Indian Rupee (Rs)
GitHub Repo	https://github.com/saisathwikgoudbodige/VIP-C2-ShopEEZ-E-Commerce-Application

2. Customer Features

- User Registration & JWT-secured Login
- Product Catalog with Search, Category, Price & Rating Filters
- Product Detail Page (images, specs, reviews, stock indicator)
- Shopping Cart with LocalStorage persistence (survives browser refresh)
- Cart Totals: Subtotal + 18% GST Tax + Shipping Fee
- Wishlist (Redux + LocalStorage)
- Multi-step Checkout with Shipping Address Form
- Order History with live Status Tracking
- Profile Management (name, address, phone, password change)
- Product Reviews & Star Ratings

3. Admin Features

- Admin Dashboard (Revenue Rs, Orders, Products, Users KPI cards)
- Low Stock Alerts (products with stock <= 5)
- Recent Orders overview (last 5 orders)
- Product CRUD (Add / Edit / Delete with multi-image upload via Multer)
- Order Status Management (processing !' shipped !' delivered !' cancelled)
- User Management (View all users + Delete non-admin users)

4. Complete API Reference

Authentication — /api/auth

Method	Endpoint	Description	Auth
POST	/api/auth/register	Register a new user	Public

POST	/api/auth/login	Login and receive JWT token	Public
GET	/api/auth/me	Get current authenticated user profile	JWT
PUT	/api/auth/update-profile	Update name, address, phone	JWT
PUT	/api/auth/change-password	Change password with current verification	JWT

Products — /api/products

Method	Endpoint	Description	Auth
GET	/api/products	Paginated catalog with search/filter/sort	Public
GET	/api/products/featured	isFeatured: true products	Public
GET	/api/products/categories	Distinct category list from DB	Public
GET	/api/products/:id	Single product full details	Public
POST	/api/products/:id/review	Submit a product review	JWT

Orders — /api/orders

Method	Endpoint	Description	Auth
POST	/api/orders	Place a new order	JWT
GET	/api/orders/my-orders	Get authenticated user's orders	JWT
GET	/api/orders/:id	Get single order details	JWT

Admin — /api/admin (All require Admin JWT)

Method	Endpoint	Description
GET	/api/admin/dashboard	Stats: products, orders, users, revenue
GET	/api/admin/products	Paginated product list (admin view)
POST	/api/admin/products	Add new product (Multer image upload)
PUT	/api/admin/products/:id	Update product details
DELETE	/api/admin/products/:id	Delete product
GET	/api/admin/orders	All orders with customer info
PUT	/api/admin/orders/:id/status	Update orderStatus field
GET	/api/admin/users	List all users (no passwords)
DELETE	/api/admin/users/:id	Delete user (non-admin only)

5. Setup & Installation

1. Clone: `git clone https://github.com/saisathwikgoubodige/VIP-C2-ShopEEZ-E-Commerce-Application.git`
2. Install dependencies: `npm install` && `cd client` && `npm install`
3. Create server/.env with MONGO_URI and JWT_SECRET values
4. Seed database: `node server/seed.js`
5. Start dev servers: `npm run dev`

Default seeded credentials:

- Admin: `admin@shopez.com` / `Admin@123`
- Customer: `user@shopez.com` / `User@123`